DATE:24/01/2026

EX-01

Windows Fundamental-1

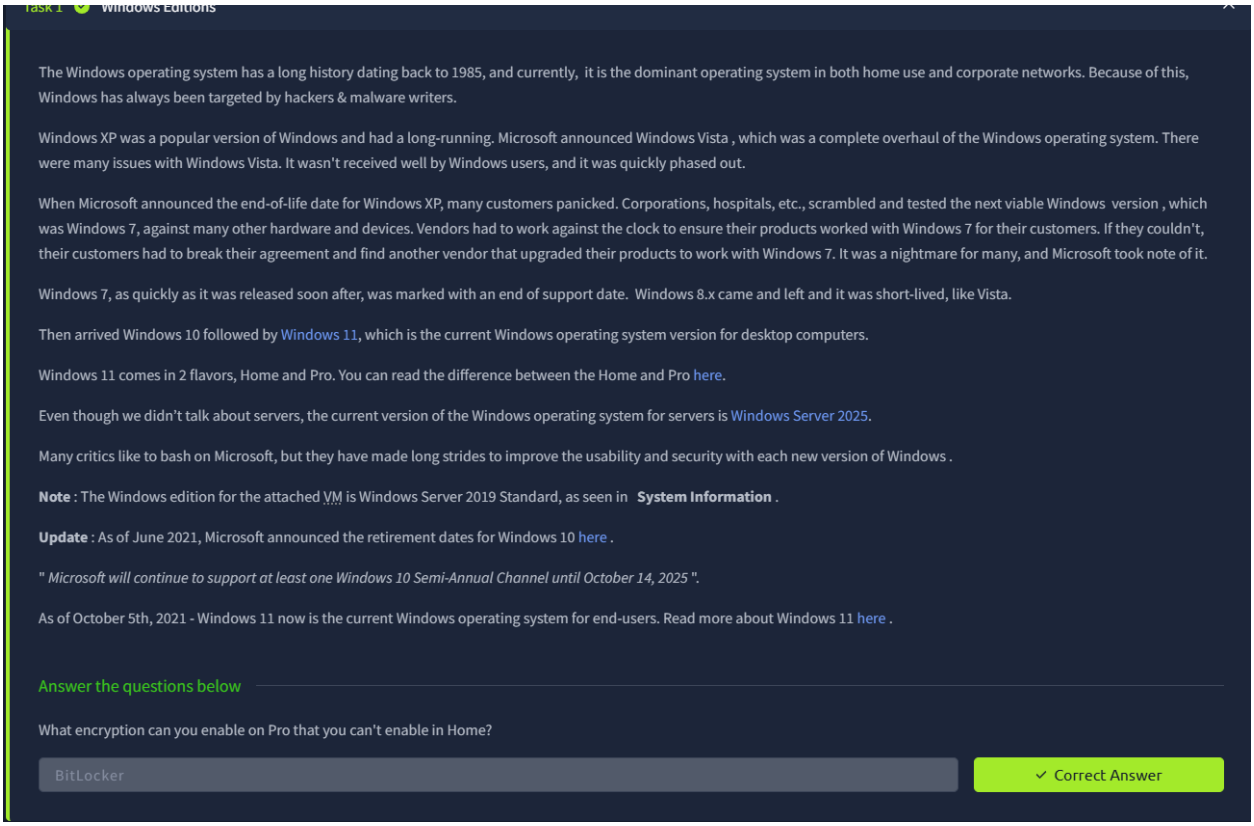
AIM:

To understand the **basic components of the Windows operating system**, including the desktop interface, NTFS file system, user accounts, and system tools such as Control Panel and Task Manager.

PROCEDURE:

- Start the virtual machine in the TryHackMe lab.
- Connect to the Windows system using Remote Desktop.
- Explore the Windows Desktop, File System (NTFS), and File Explorer.
- Open Control Panel and Settings to view system configurations.
- Observe User Account Control (UAC) and basic system tools like Task Manager.

TASK-1:



Task 1 Windows Editions

The Windows operating system has a long history dating back to 1985, and currently, it is the dominant operating system in both home use and corporate networks. Because of this, Windows has always been targeted by hackers & malware writers.

Windows XP was a popular version of Windows and had a long-running. Microsoft announced Windows Vista , which was a complete overhaul of the Windows operating system. There were many issues with Windows Vista. It wasn't received well by Windows users, and it was quickly phased out.

When Microsoft announced the end-of-life date for Windows XP, many customers panicked. Corporations, hospitals, etc., scrambled and tested the next viable Windows version , which was Windows 7, against many other hardware and devices. Vendors had to work against the clock to ensure their products worked with Windows 7 for their customers. If they couldn't, their customers had to break their agreement and find another vendor that upgraded their products to work with Windows 7. It was a nightmare for many, and Microsoft took note of it.

Windows 7, as quickly as it was released soon after, was marked with an end of support date. Windows 8.x came and left and it was short-lived, like Vista.

Then arrived Windows 10 followed by Windows 11, which is the current Windows operating system version for desktop computers.

Windows 11 comes in 2 flavors, Home and Pro. You can read the difference between the Home and Pro here.

Even though we didn't talk about servers, the current version of the Windows operating system for servers is Windows Server 2025.

Many critics like to bash on Microsoft, but they have made long strides to improve the usability and security with each new version of Windows .

Note : The Windows edition for the attached VM is Windows Server 2019 Standard, as seen in **System Information** .

Update : As of June 2021, Microsoft announced the retirement dates for Windows 10 here .

" Microsoft will continue to support at least one Windows 10 Semi-Annual Channel until October 14, 2025 ".

As of October 5th, 2021 - Windows 11 now is the current Windows operating system for end-users. Read more about Windows 11 here .

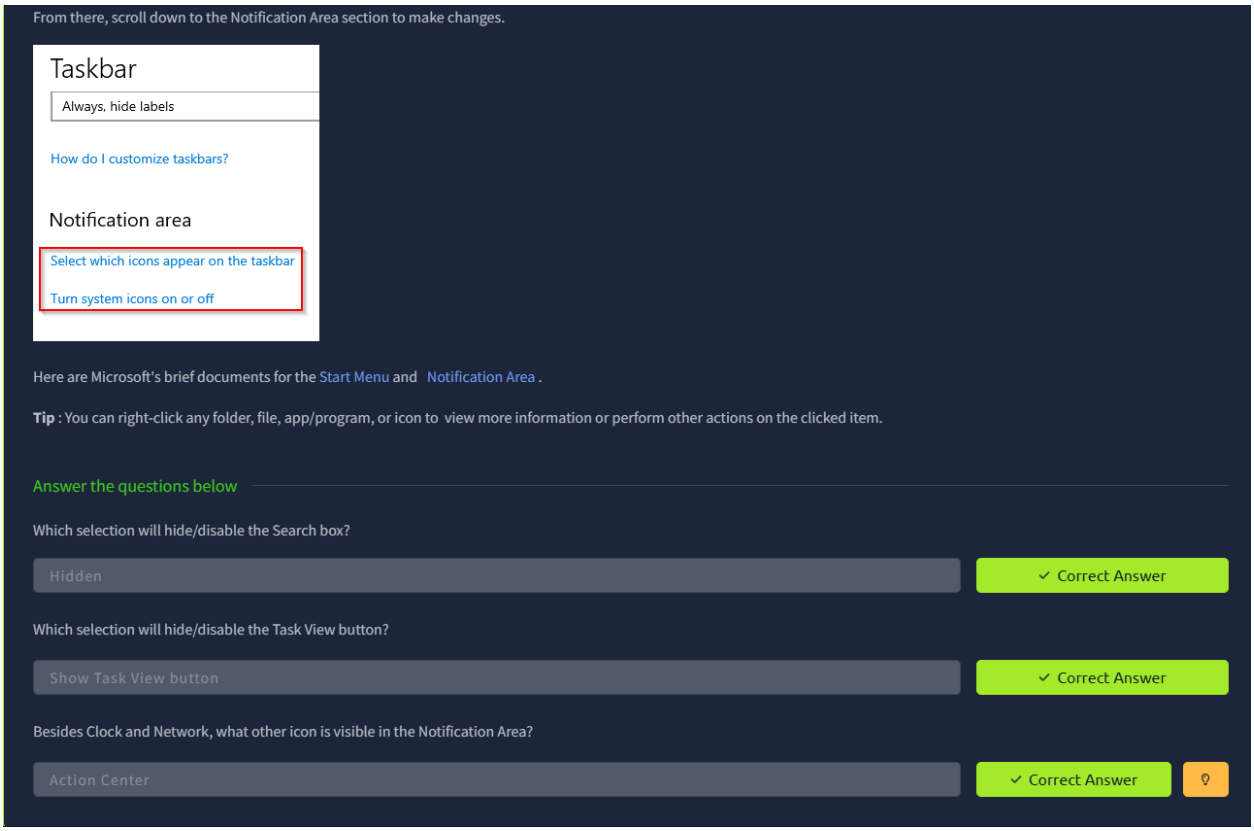
Answer the questions below

What encryption can you enable on Pro that you can't enable in Home?

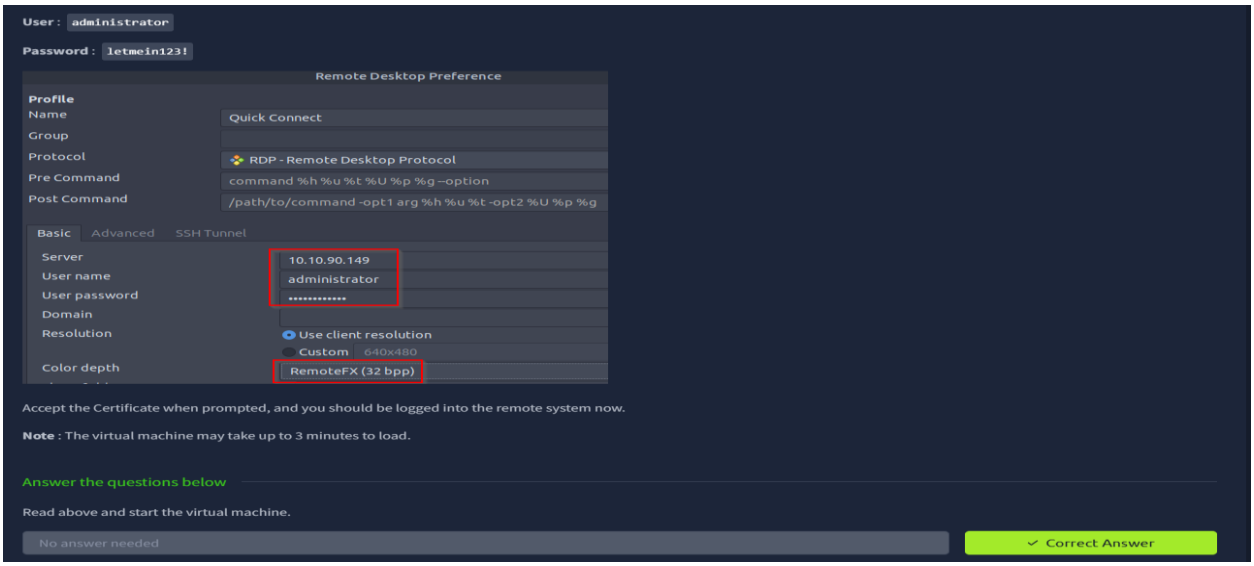
BitLocker

✓ Correct Answer

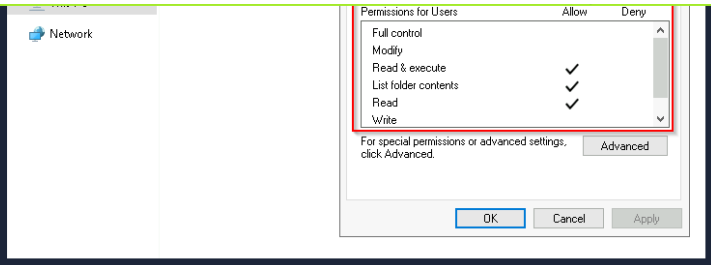
TASK-2:


From there, scroll down to the Notification Area section to make changes.
Taskbar
Always, hide labels
How do I customize taskbars?
Notification area
Select which icons appear on the taskbar
Turn system icons on or off
Here are Microsoft's brief documents for the Start Menu and Notification Area .
Tip : You can right-click any folder, file, app/program, or icon to view more information or perform other actions on the clicked item.
Answer the questions below
Which selection will hide/disable the Search box?
Hidden
Correct Answer
Which selection will hide/disable the Task View button?
Show Task View button
Correct Answer
Besides Clock and Network, what other icon is visible in the Notification Area?
Action Center
Correct Answer

TASK-3:


User : administrator
Password : letmein123!
Remote Desktop Preference
Profile
Name
Quick Connect
Group
Protocol
RDP - Remote Desktop Protocol
Pre Command
command %h %u %t %U %p %g --option
Post Command
/path/to/command -opt1 arg %h %u %t -opt2 %U %p %g
Basic
Advanced
SSH Tunnel
Server
10.10.90.149
User name
administrator
User password
Domain
Resolution
Use client resolution
Custom
640x480
Color depth
RemoteFX (32 bpp)
Accept the Certificate when prompted, and you should be logged into the remote system now.
Note : The virtual machine may take up to 3 minutes to load.
Answer the questions below
Read above and start the virtual machine.
No answer needed
Correct Answer

TASK-4:



Refer to the Microsoft documentation to get a better understanding of the NTFS permissions for Special Permissions .

Another feature of NTFS is **Alternate Data Streams** (**ADS**).

Alternate Data Streams (ADS) is a file attribute specific to Windows **NTFS** (New Technology File System).

Every file has at least one data stream (`$DATA`), and ADS allows files to contain more than one stream of data. Natively Window Explorer doesn't display ADS to the user. There are 3rd party executables that can be used to view this data, PowerShell also gives you the ability to view ADS for files. We will cover how you can use PowerShell to view any ADS for any files in the Windows PowerShell room.

From a security perspective, malware writers have used ADS to hide data.

Not all its uses are malicious. For example, when you download a file from the Internet, there are identifiers written to ADS to identify that the file was downloaded from the Internet.

To learn more about ADS, refer to the following link from MalwareBytes here .

Bonus : If you wish to interact hands-on with ADS, I suggest exploring Day 21 of Advent of Cyber 2 .

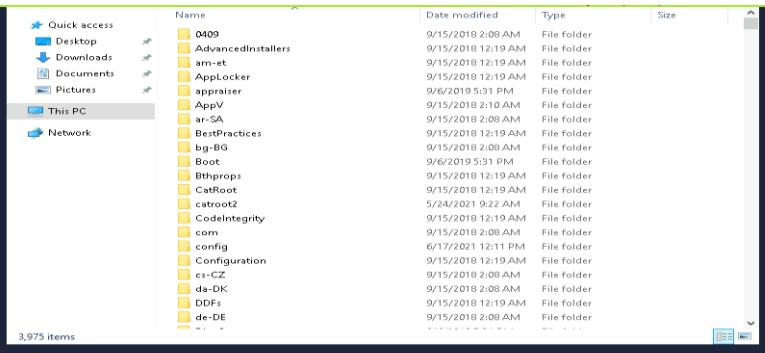
Answer the questions below

What is the meaning of NTFS?

New Technology File System

✓ Correct Answer

TASK-5:



The System32 folder holds the important files that are critical for the operating system.

You should proceed with extreme caution when interacting with this folder. Accidentally deleting any files or folders within System32 can render the Windows OS inoperational. Read more about this action here .

Note : Many of the tools that will be covered in the Windows Fundamentals series reside within the System32 folder.

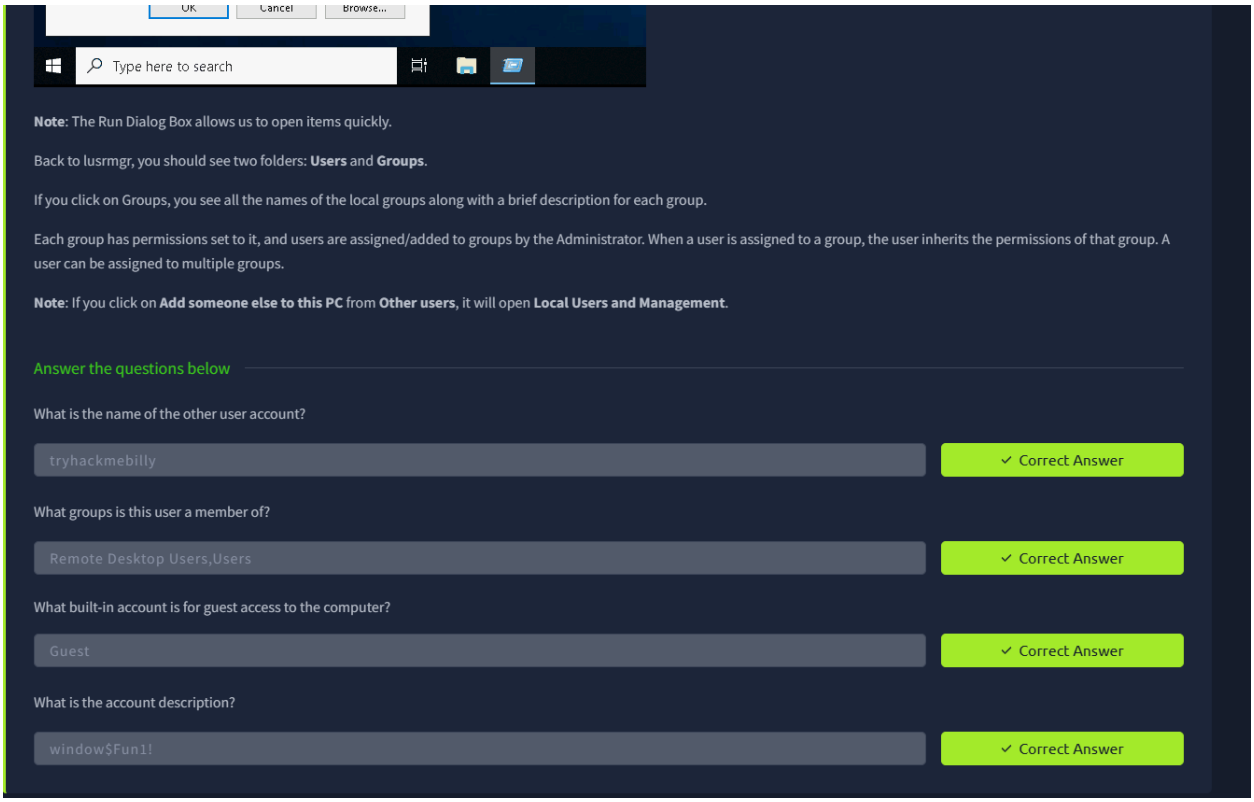
Answer the questions below

What is the system variable for the Windows folder?

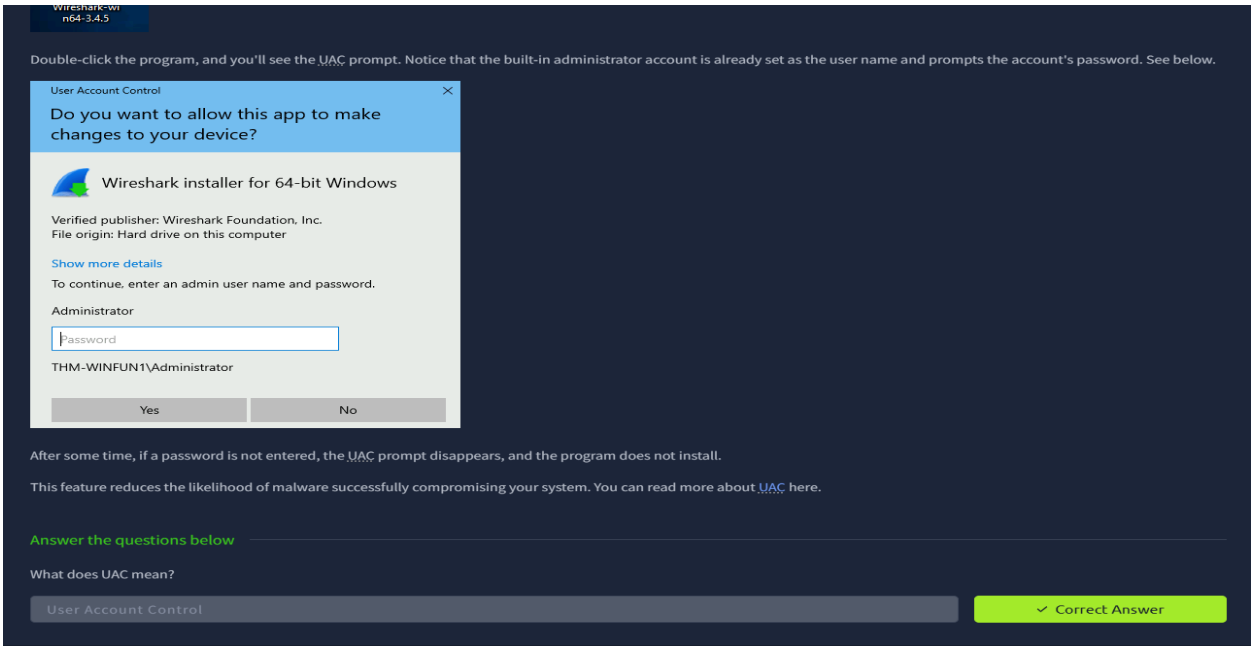
%windir%

✓ Correct Answer

TASK-6:

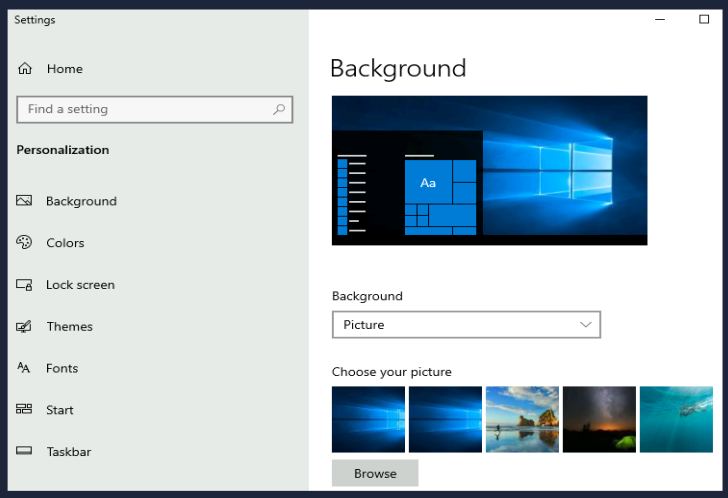

OK
Cancel
Browse...
Type here to search
Note: The Run Dialog Box allows us to open items quickly.
Back to lusrmgr, you should see two folders: Users and Groups.
If you click on Groups, you see all the names of the local groups along with a brief description for each group.
Each group has permissions set to it, and users are assigned/added to groups by the Administrator. When a user is assigned to a group, the user inherits the permissions of that group. A user can be assigned to multiple groups.
Note: If you click on Add someone else to this PC from Other users, it will open Local Users and Management.
Answer the questions below
What is the name of the other user account?
tryhackmebilly
Correct Answer
What groups is this user a member of?
Remote Desktop Users,Users
Correct Answer
What built-in account is for guest access to the computer?
Guest
Correct Answer
What is the account description?
window$Fun1!
Correct Answer

TASK-7:


n64-3.4.5
Double-click the program, and you'll see the UAC prompt. Notice that the built-in administrator account is already set as the user name and prompts the account's password. See below.
User Account Control
Do you want to allow this app to make changes to your device?
Wireshark installer for 64-bit Windows
Verified publisher: Wireshark Foundation, Inc.
File origin: Hard drive on this computer
Show more details
To continue, enter an admin user name and password.
Administrator
Password
THM-WINFUN1\Administrator
Yes
No
After some time, if a password is not entered, the UAC prompt disappears, and the program does not install.
This feature reduces the likelihood of malware successfully compromising your system. You can read more about UAC here.
Answer the questions below
What does UAC mean?
User Account Control
Correct Answer

TASK-8:

If we click on the Best match, a window to the Settings menu appears to make changes to the wallpaper.



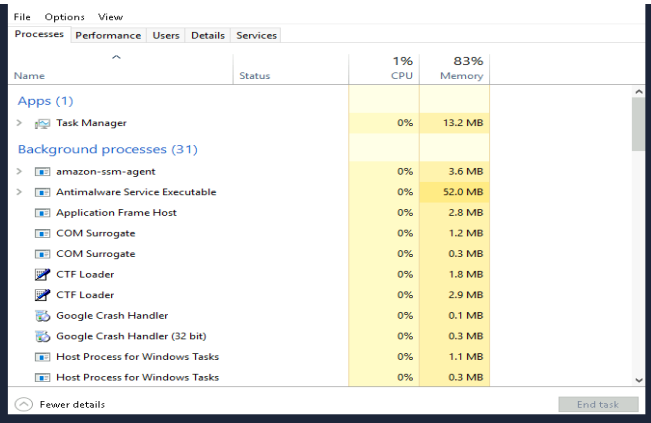
Answer the questions below

In the Control Panel, change the view to **Small icons**. What is the last setting in the Control Panel view?

Windows Defender Firewall

✓ Correct Answer

TASK-9:



You can refer to this blog post for more detailed information about the Task Manager.

If you wish to learn more about the core Windows processes and what each process is responsible for, visit the Core Windows Processes room.

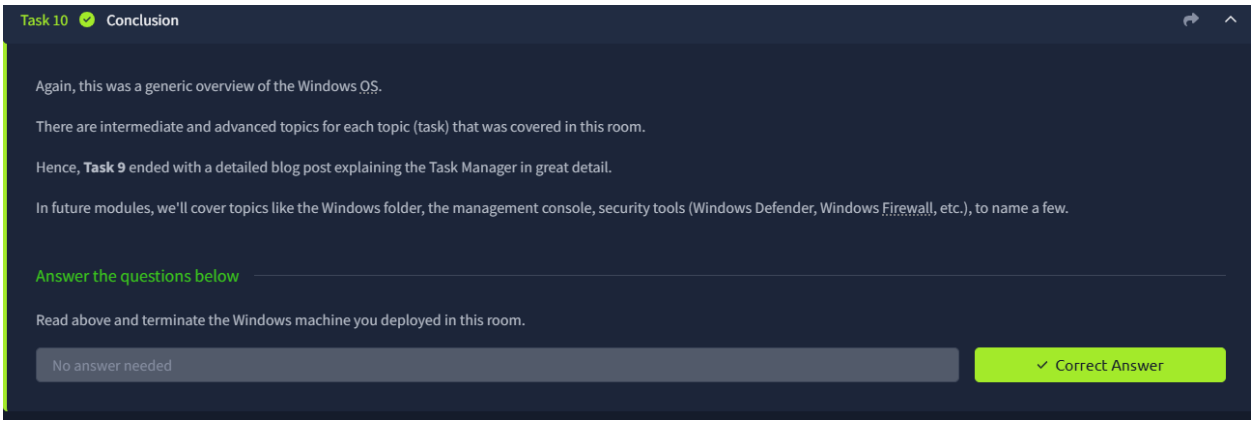
Answer the questions below

What is the keyboard shortcut to open Task Manager?

Ctrl+Shift+Esc

✓ Correct Answer

TASK-10:



RESULT:

Successfully learned the fundamentals of windows od through Task manager.

Windows fundamentals-2

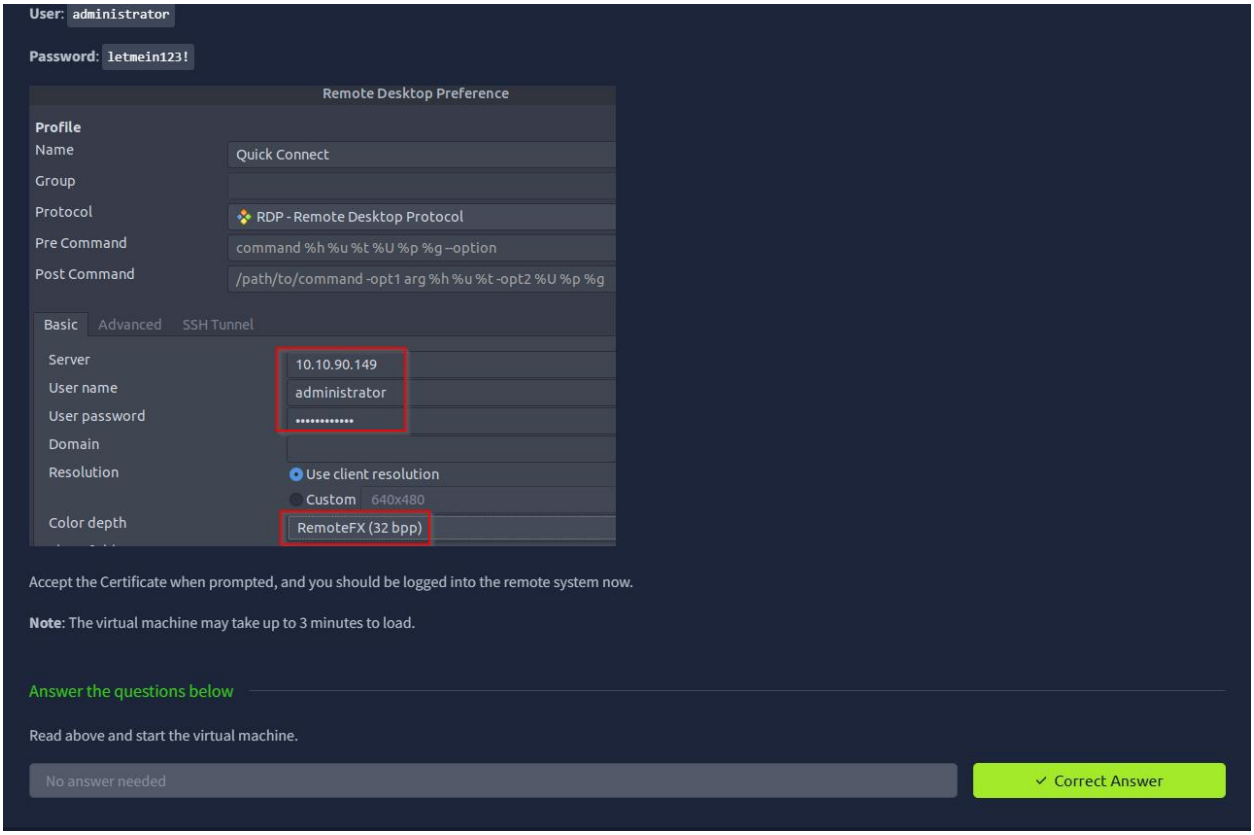
AIM:

To study **Windows administrative utilities and system management tools** such as System Configuration, Computer Management, Resource Monitor, and the Windows Registry.

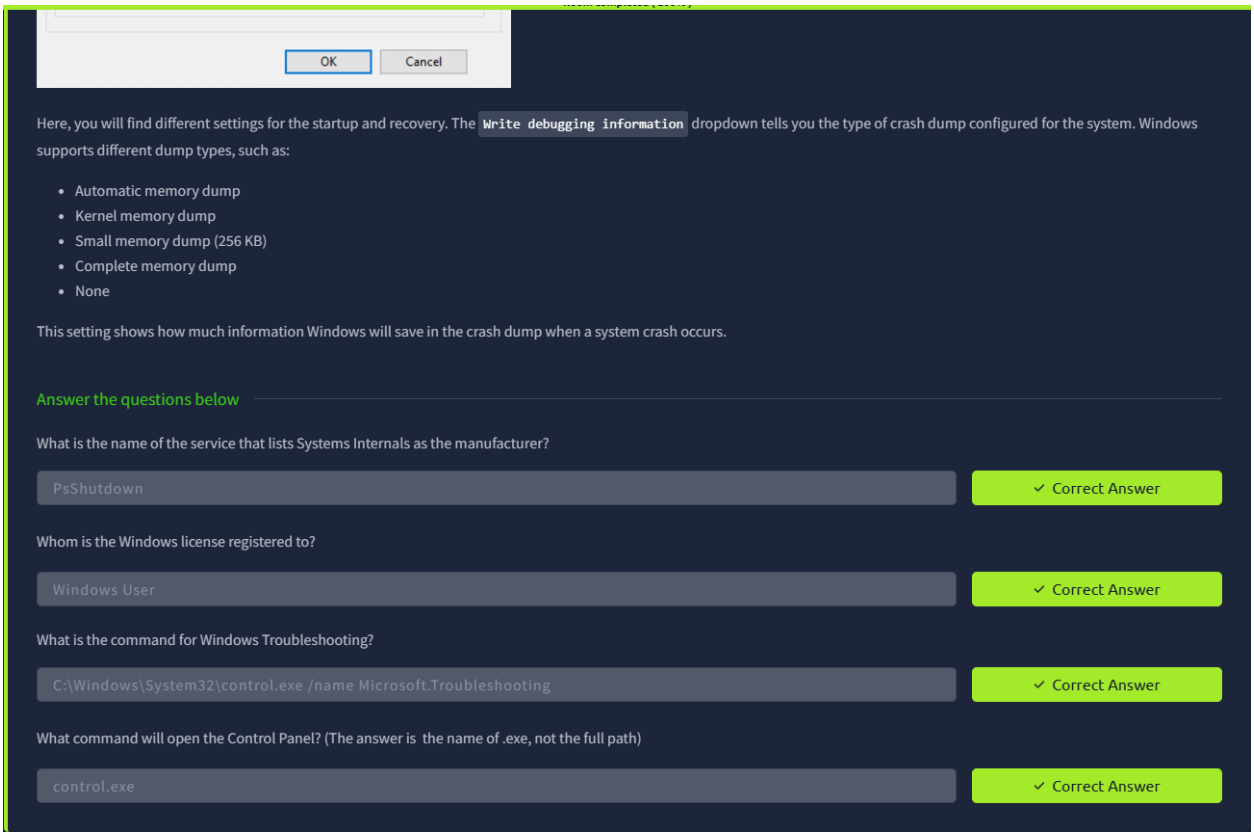
PROCEDURE:

- Launch the lab virtual machine.
- Connect to the system using the provided credentials.
- Open **System Configuration and Computer Management** tools.
- Explore utilities like **Task Scheduler** and **Event Viewer**.
- Use **Resource Monitor** to analyze CPU, disk, memory, and network usage.
- Review logs and system processes and complete the lab questions.

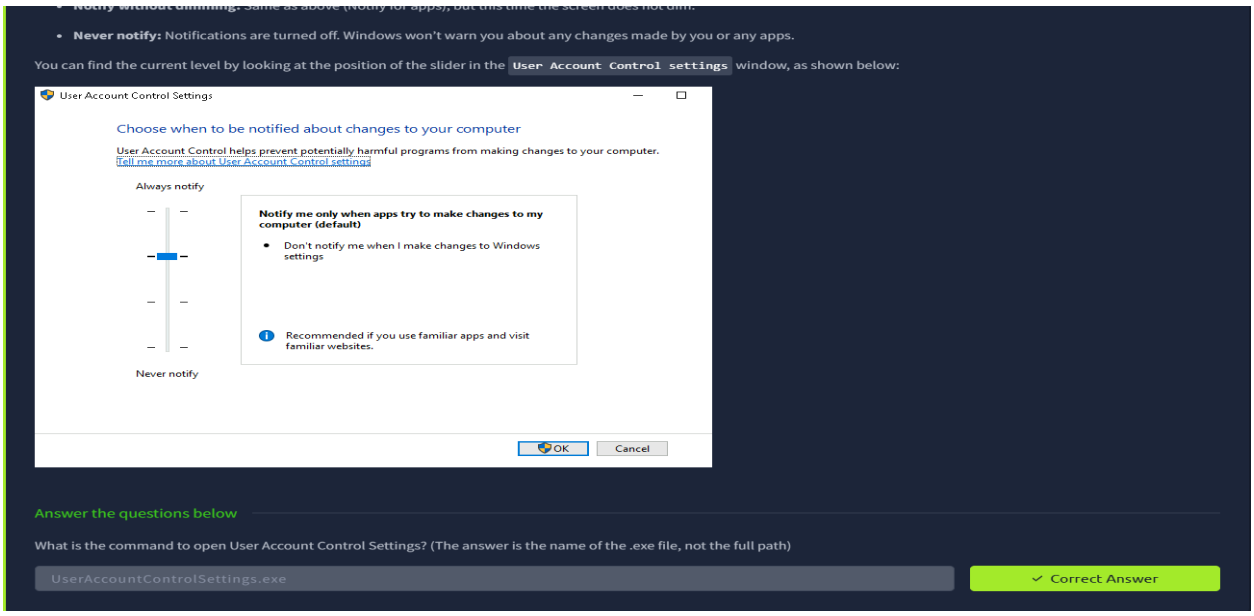
TASK-1:



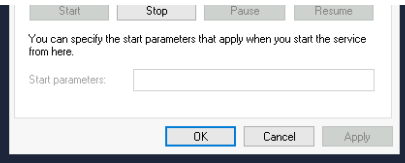
TASK-2:


OK
Cancel
Here, you will find different settings for the startup and recovery. The Write debugging information dropdown tells you the type of crash dump configured for the system. Windows supports different dump types, such as:
Automatic memory dump
Kernel memory dump
Small memory dump (256 KB)
Complete memory dump
None
This setting shows how much information Windows will save in the crash dump when a system crash occurs.
Answer the questions below
What is the name of the service that lists Systems Internals as the manufacturer?
PsShutdown
Correct Answer
Whom is the Windows license registered to?
Windows User
Correct Answer
What is the command for Windows Troubleshooting?
C:\Windows\System32\control.exe /name Microsoft.Troubleshooting
Correct Answer
What command will open the Control Panel? (The answer is the name of .exe, not the full path)
control.exe
Correct Answer

TASK-3:


Never notify: Notifications are turned off. Windows won't warn you about any changes made by you or any apps.
You can find the current level by looking at the position of the slider in the User Account Control settings window, as shown below:
User Account Control Settings
Choose when to be notified about changes to your computer
User Account Control helps prevent potentially harmful programs from making changes to your computer.
Tell me more about User Account Control settings
Always notify
Notify me only when apps try to make changes to my computer (default)
Don't notify me when I make changes to Windows settings
Recommended if you use familiar apps and visit familiar websites.
Never notify
OK
Cancel
Answer the questions below
What is the command to open User Account Control Settings? (The answer is the name of the .exe file, not the full path)
UserAccountControlSettings.exe
Correct Answer

TASK-4:



There is a field known as Startup type in a service's Properties window, as shown above. It determines how and when the service is configured to start. We can set a service to `Automatic`, which means it starts every time the system boots, or `Manual`, which means it only starts when another process or user triggers this service, or `Disabled`, which means it should not run at all. The service shown in the screenshot above is set to `Automatic`.

WMI Control configures and controls the **Windows Management Instrumentation** (WMI) service.

Per Wikipedia, "*WMI allows scripting languages (such as VBScript or Windows PowerShell) to manage Microsoft Windows personal computers and servers, both locally and remotely. Microsoft also provides a command-line interface to WMI called Windows Management Instrumentation Command-line (WMIC).*"

Note: The WMIC tool is deprecated in Windows 10, version 21H1. Windows PowerShell supersedes this tool for WMI.

Answer the questions below

What is the command to open Computer Management?
(The answer is the name of the .msc file, not the full path)

compmgmt.msc — ✓ Correct Answer

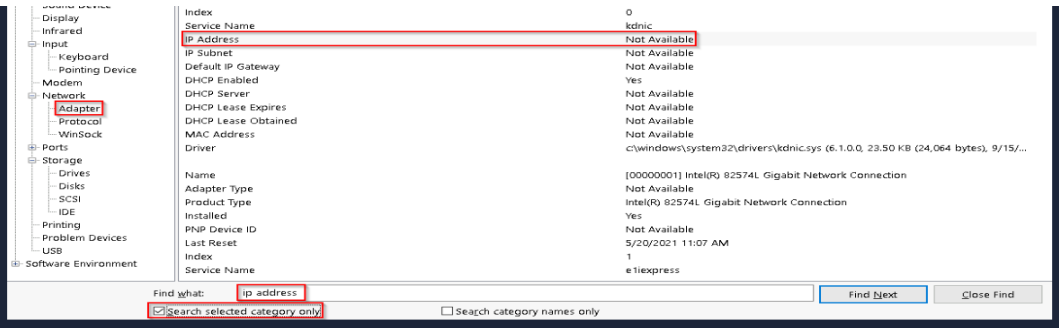
When is the `npcapwatchdog` scheduled task set to run at?

At system startup — ✓ Correct Answer

What is the name of the hidden folder that is shared?

sh4r3dF0Ld3r — ✓ Correct Answer

TASK-5:



Answer the questions below

What is the command to open System Information? (The answer is the name of the .exe file, not the full path)

msinfo32.exe — ✓ Correct Answer

What is listed under System Name?

THM-WINFUN2 — ✓ Correct Answer

Under Environment Variables, what is the value for ComSpec?

%SystemRoot%\system32\cmd.exe — ✓ Correct Answer

TASK-6:

System	4	C:\Windows\System32\winevt\Logs\Microsoft-Windows-SystemDataArchiver%4Diagnostic....	0	1,416	1,416	Normal	1
System	4	C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTDefenderApiLogger.etl	0	675	675	Normal	1
System	4	C:\$Extend\$UsnJrnl:$J	0	455	455	Normal	1
System	4	C:\$Mft (NTFS Master File Table)	0	801	801	Normal	1
System	4	C:\$BitMap (NTFS Free Space Map)	0	819	819	Normal	1

Storage

Logical Disk	Physical Disk	Active Time (%)	Available Space...	Total Space (MB)	Disk Queue Le...
C:	0	0.00	9,304	19,929	0.00

Network

Overview | CPU | Memory | Disk | Network

Processes with Network Activity

Image	PID	Send (B/sec)	Receive (B/sec)	Total (B/sec)
svchost.exe (termsvcs)	528	7,522	1,035	8,557

Network Activity — 29 Kbps Network I/O — 0% Network Utilization

Image	PID	Address	Send (B/sec)	Receive (B/sec)	Total (B/sec)
svchost.exe (termsvcs)	528	ip-10-13-2-191.eu-west-1.compute.internal	7,522	1,035	8,557

TCP Connections

Image	PID	Local Address	Local Port	Remote Address	Remote Port	Packet Loss (%)	Latency (ms)
svchost.exe (termsvcs)	528	10.10.197.248	3389	10.13.2.191	52350	0	234

Listening Ports

Image	PID	Address	Port	Protocol	Firewall Status
svchost.exe (LocalService)	1220	IPv6 unspecified	123	UDP	Allowed, restri...

Although not captured in any of the images above, Resource Monitor has a pane at the far right. This pane shows a graphical view in real-time for each section.

Note: The information displayed in Resource Monitor will be different for you compared to the images above.

Answer the questions below

What is the command to open Resource Monitor? (The answer is the name of the .exe file, not the full path)

resmon.exe

✓ Correct Answer

TASK-7:

```
NET CONTINUE                NET SESSION                 NET USE
NET FILE                    NET SHARE                   NET USER
NET GROUP                   NET START                   NET VIEW
NET HELP

NET HELP NAMES explains different types of names in NET HELP syntax lines.
NET HELP SERVICES lists some of the services you can start.
NET HELP SYNTAX explains how to read NET HELP syntax lines.
NET HELP command | MORE displays Help one screen at a time.
```

So, if you wish to see the help information for `net user`, the command is `net help user`.

```
C:\Users\Administrator>net help user
The syntax of this command is:

NET USER
[username [password | *] [options]] [/DOMAIN]
         username {password | *} /ADD [options] [/DOMAIN]
         username [/DELETE] [/DOMAIN]
         username [/TIMES:{times | ALL}]
         username [/ACTIVE: {YES | NO}]

NET USER creates and modifies user accounts on computers. When used
without switches, it lists the user accounts for the computer. The
user account information is stored in the user accounts database.
```

You can use the same command to view the help information for other useful **net** sub-commands, such as **localgroup**, **use**, **share**, and **session**.

Refer to the following link to see a comprehensive list of commands you can execute in the command prompt here.

Answer the questions below

In System Configuration, what is the full command for Internet Protocol Configuration?

C:\Windows\System32\cmd.exe /k %windir%\system32\ipconfig.exe

✓ Correct Answer

For the ipconfig command, how do you show detailed information?

ipconfig /all

✓ Correct Answer

TASK-8:

We're continuing with Tools that are available through the **System Configuration** panel.

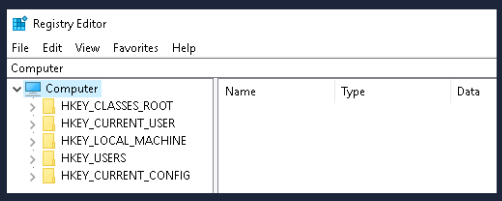
The **Windows Registry** (per Microsoft) is a central hierarchical database used to store information necessary to configure the system for one or more users, applications, and hardware devices.

The registry contains information that Windows continually references during operation, such as:

- Profiles for each user
- Applications installed on the computer and the types of documents that each can create
- Property sheet settings for folders and application icons
- What hardware exists on the system
- The ports that are being used.

Warning: The registry is for advanced computer users. Making changes to the registry can affect normal computer operations.

There are various ways to view/edit the registry. One way is to use the **Registry Editor** (`regedit`).



Refer to the following Microsoft documentation here to learn more about the Windows Registry.

Answer the questions below

What is the command to open the Registry Editor? (The answer is the name of the .exe file, not the full path)

regedt32.exe

✓ Correct Answer

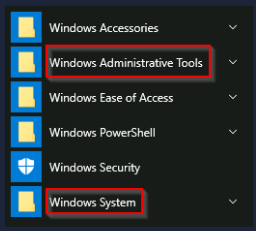
TASK-9:

Task 9 Conclusion

Recall that the tasks covered in this room were some of the tools that can launch from `MSConfig`.

Throughout the room, commands and shortcuts were shared for the utilities. This means you don't have to launch **MSConfig** to run these utilities.

You can also run some of these utilities directly from the Start Menu. See below where some of these utilities can be found.



Some of the tools listed in **MSConfig** that weren't mentioned in this room were either covered in Windows Fundamentals 1 or were left for you to explore on your own.

Answer the questions below

Read above.

No answer needed

✓ Correct Answer

RESULT:

Successfully learned the fundamentals of windows through Task manager.

Windows Fundamentals-3

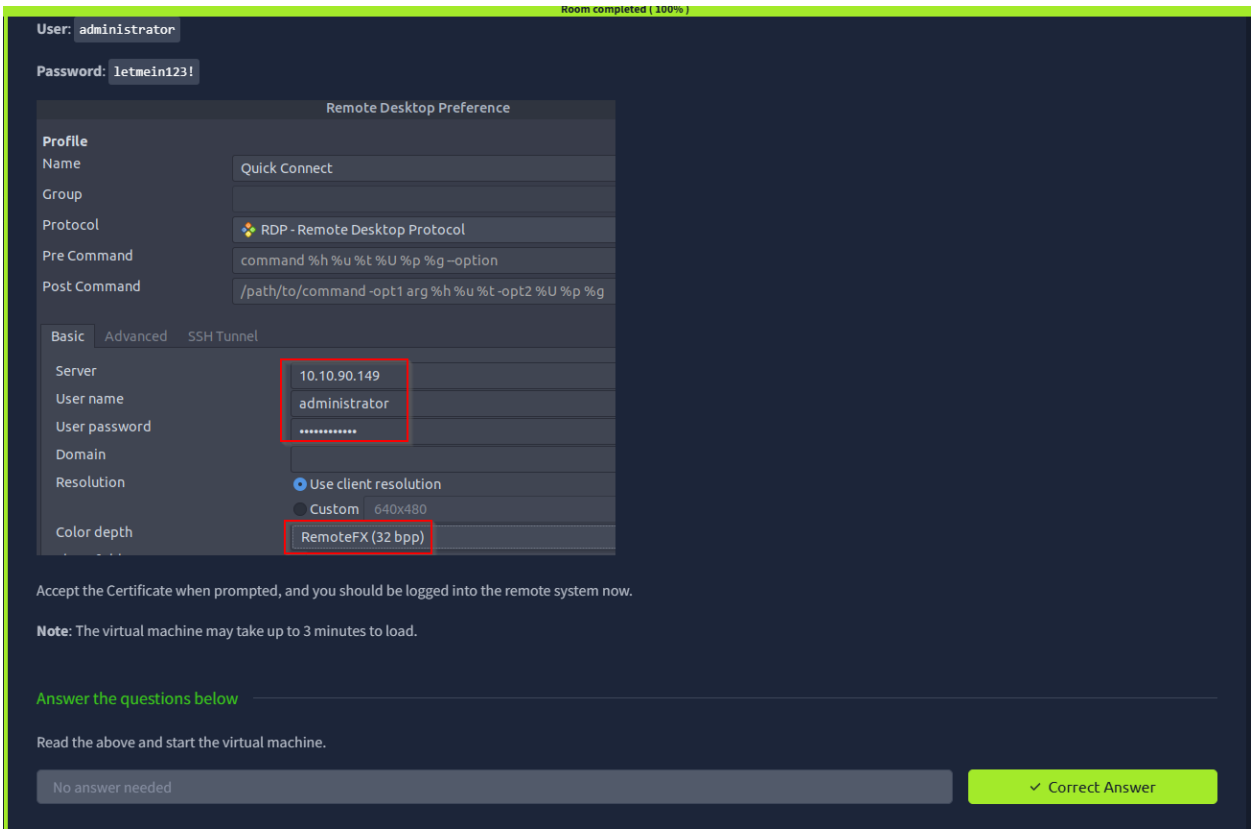
AIM:

To explore **Windows security features** including Windows Update, Microsoft Defender, Firewall, BitLocker, and other tools used to protect the operating system.

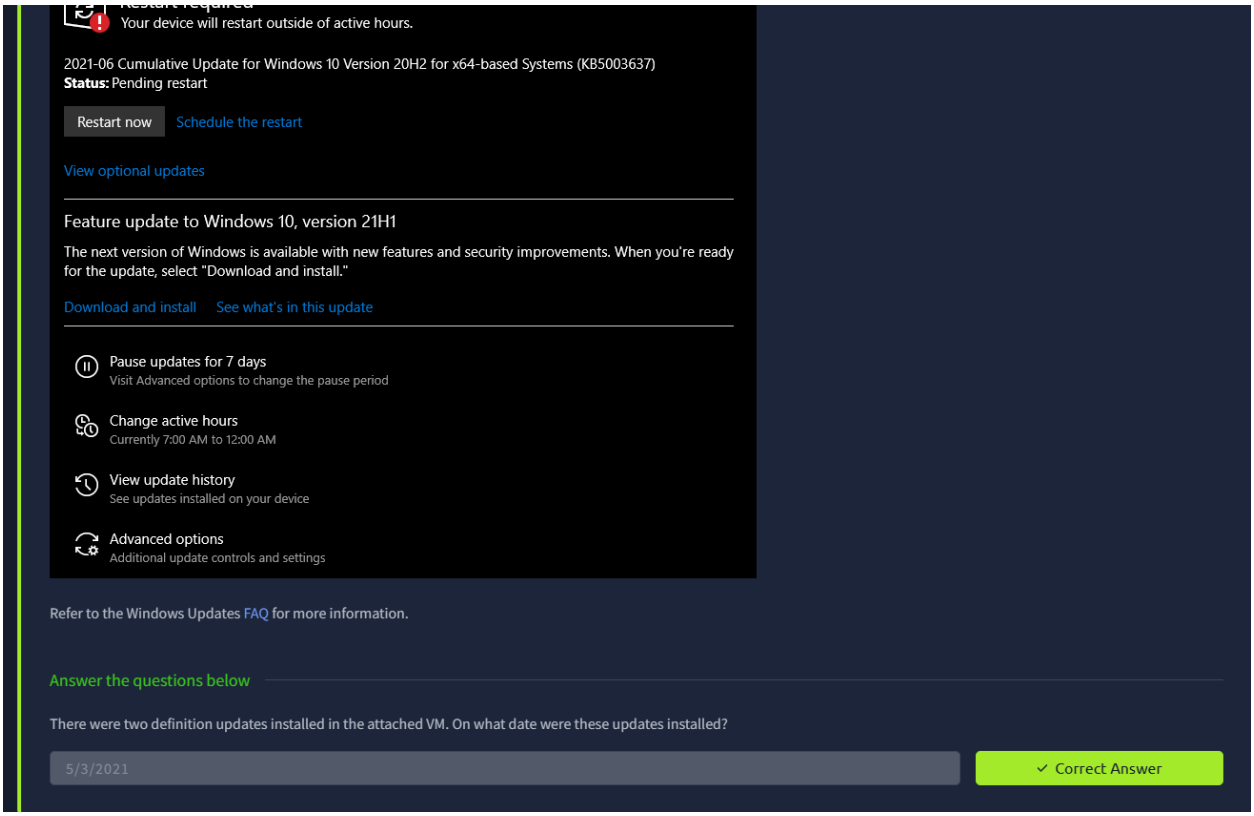
PROCEDURE:

- Start the virtual machine and log in via Remote Desktop.
- Open **Windows Security settings**.
- Explore security features such as **Windows Update**, **Microsoft Defender**, and **Firewall**.
- Examine **BitLocker encryption and system protection features**.
- Study **Volume Shadow Copy Service (VSS)** for backup and recovery.

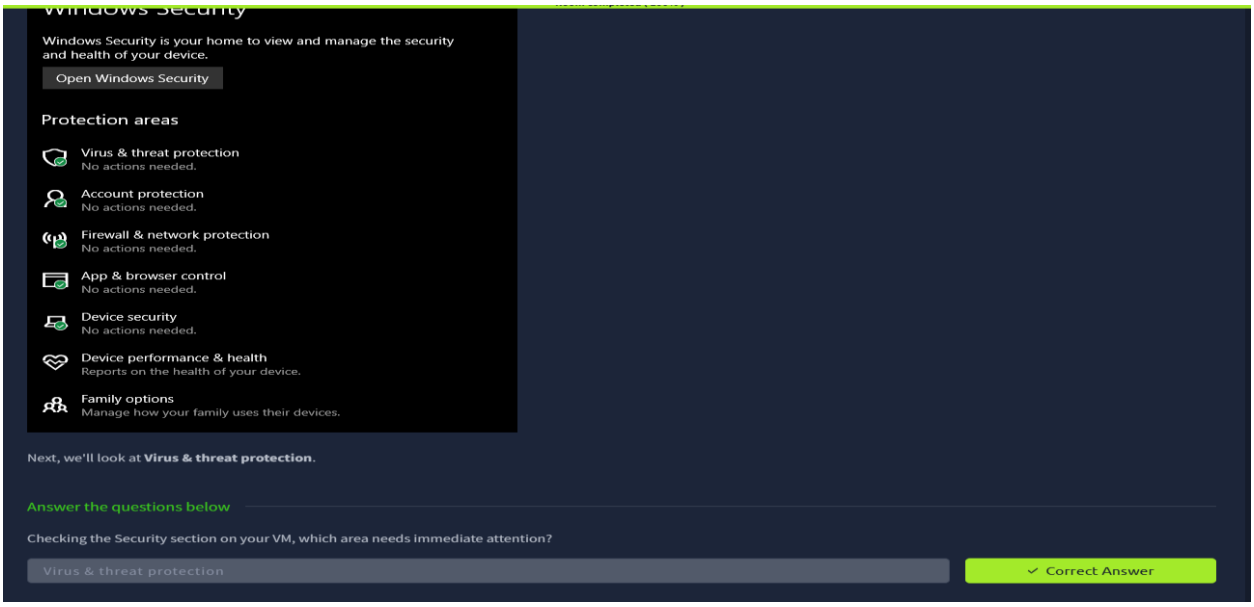
TASK-1:



TASK-2:



TASK-3:



TASK-4:

is enabled, only approved and trusted apps would be allowed to modify the files in the protected folders. To enable this feature, click on the `Manage controlled folder access` button under `Controlled Folder Access` and turn it on.

- **Exclusions** - Windows Defender Antivirus allows you to exclude any files or folders from the antivirus scanning. This is done to reduce the number of false positives. Administrators might not want the antivirus to scan specific files or folders. By adding them to the exclusions list, the antivirus would ignore them and scan all the other files and folders. To add any file or folder to the Windows Defender exclusion list, click on the `Add or remove exclusions` button under `Exclusions` and add as many exclusions as you want.
- **Notifications** - Windows Defender Antivirus will send notifications with critical information about the health and security of your device.

Warning: Excluded items could contain threats that make your device vulnerable. Only use this option if you are **100%** sure of what you are doing.

Virus & threat protection updates

- **Check for updates** - Manually check for updates to update Windows Defender Antivirus definitions.

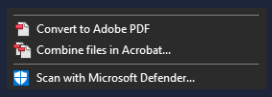
Ransomware protection

- **Controlled folder access** - Ransomware protection requires this feature to be enabled, which in turn requires Real-time protection to be enabled.

Note: Real-time protection is turned off in the attached VM to decrease the chances of performance issues. Since the VM can't reach the Internet and there aren't any threats in the VM, this is safe to do. Real-time protection should definitely be enabled in your personal Windows devices unless you have a 3rd party product that provides the same protection. Ensure it's always up-to-date and enabled.

Tip: You can perform on-demand scans on any file/folder by right-clicking the item and selecting 'Scan with Microsoft Defender'.

The below image was taken from another Windows device to show this feature.



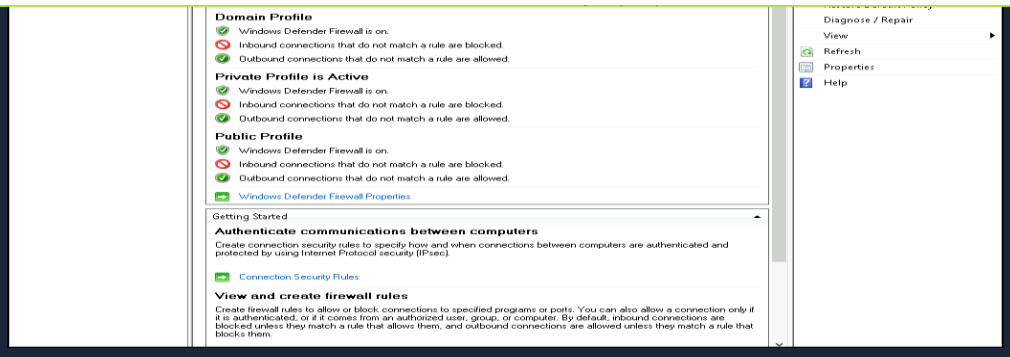
Answer the questions below

Specifically, what is turned off that Windows is notifying you to turn on?

Real-time protection

✓ Correct Answer

TASK-5:



Configuring the **Windows Defender Firewall** is for advanced Windows users. Refer to the following Microsoft documentation on best practices here.

Tip: Command to open the Windows Defender Firewall is `WF.msc`.

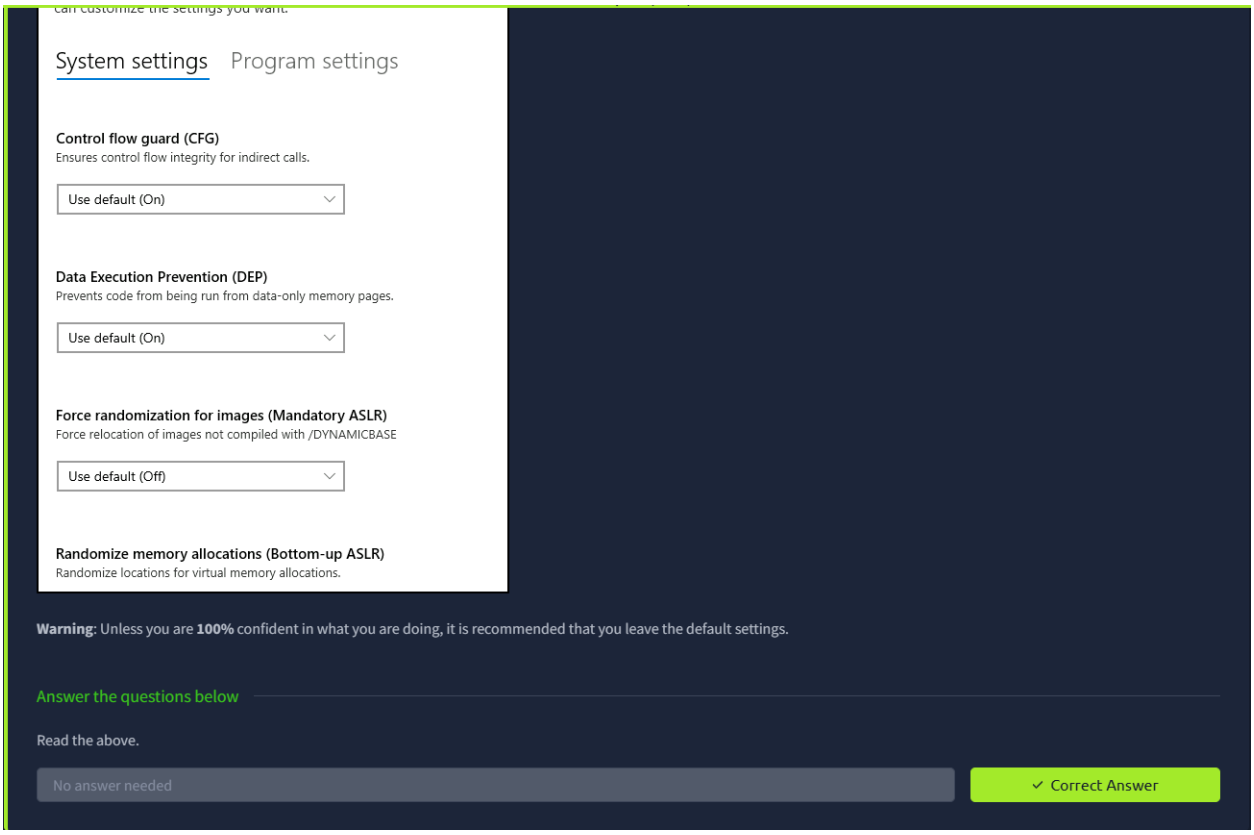
Answer the questions below

If you were connected to airport Wi-Fi, what most likely will be the active firewall profile?

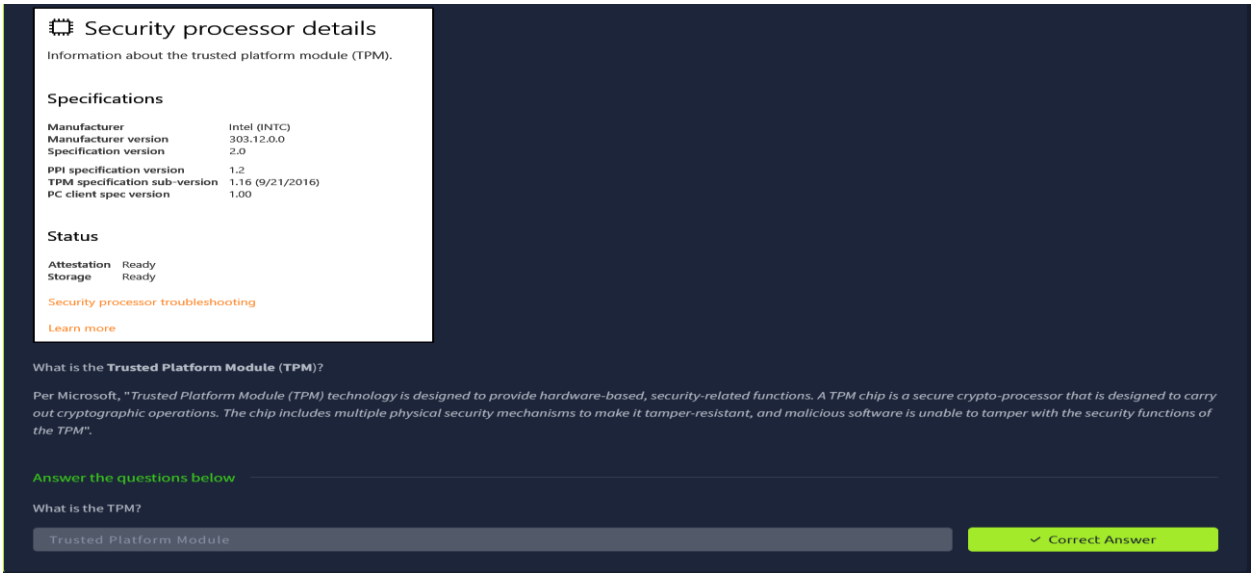
Public network

✓ Correct Answer

TASK-6:


can customize the settings you want.
System settings
Program settings
Control flow guard (CFG)
Ensures control flow integrity for indirect calls.
Use default (On)
Data Execution Prevention (DEP)
Prevents code from being run from data-only memory pages.
Use default (On)
Force randomization for images (Mandatory ASLR)
Force relocation of images not compiled with /DYNAMICBASE
Use default (Off)
Randomize memory allocations (Bottom-up ASLR)
Randomize locations for virtual memory allocations.
Warning: Unless you are 100% confident in what you are doing, it is recommended that you leave the default settings.
Answer the questions below
Read the above.
No answer needed
✓ Correct Answer

TASK-7:


Security processor details
Information about the trusted platform module (TPM).
Specifications
Manufacturer Intel (INTC)
Manufacturer version 303.12.0.0
Specification version 2.0
PPI specification version 1.2
TPM specification sub-version 1.16 (9/21/2016)
PC client spec version 1.00
Status
Attestation Ready
Storage Ready
Security processor troubleshooting
Learn more
What is the Trusted Platform Module (TPM)?
Per Microsoft, "Trusted Platform Module (TPM) technology is designed to provide hardware-based, security-related functions. A TPM chip is a secure crypto-processor that is designed to carry out cryptographic operations. The chip includes multiple physical security mechanisms to make it tamper-resistant, and malicious software is unable to tamper with the security functions of the TPM".
Answer the questions below
What is the TPM?
Trusted Platform Module
✓ Correct Answer

TASK-8:

Task 8 BitLocker

What is **BitLocker**?

Per Microsoft, "*BitLocker Drive Encryption is a data protection feature that integrates with the operating system and addresses the threats of data theft or exposure from lost, stolen, or inappropriately decommissioned computers*".

On devices with TPM installed, BitLocker offers the best protection.

Per Microsoft, "*BitLocker provides the most protection when used with a Trusted Platform Module (TPM) version 1.2 or later. The TPM is a hardware component installed in many newer computers by the computer manufacturers. It works with BitLocker to help protect user data and to ensure that a computer has not been tampered with while the system was offline*".

Refer to the official Microsoft documentation to learn more about BitLocker here.

Note: The BitLocker feature is not included in the attached VM.

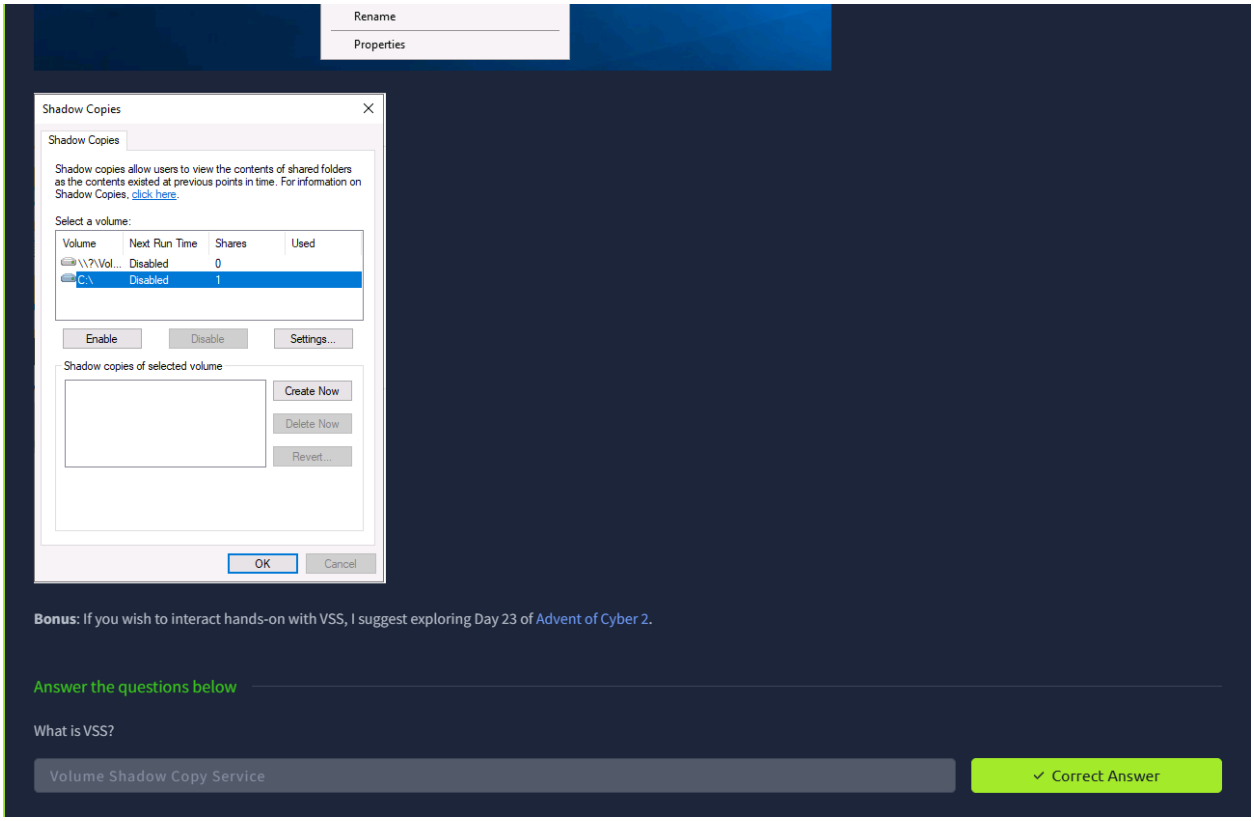
Answer the questions below

We should use a removable drive on systems **without** a TPM version 1.2 or later. What does this removable drive contain?

startup key

✓ Correct Answer

TASK-9:



Bonus: If you wish to interact hands-on with VSS, I suggest exploring Day 23 of Advent of Cyber 2.

Answer the questions below

What is VSS?

Volume Shadow Copy Service

✓ Correct Answer

TASK-10:

Task 10 Conclusion

In this room, we covered several built-in Windows security tools that ship with the Windows OS to help keep the device protected.

There is still so much to explain and cover regarding the Windows OS. As mentioned in the Windows Fundamentals 1 room, "*The content is aimed at those who wish to understand and use the Windows OS on a more comfortable level* ."

To learn more about the Windows OS, you'll need to continue the journey on your own.

Further reading material:

- Antimalware Scan Interface
- Credential Guard
- Windows 10 Hello
- CSO Online - The best new Windows 10 security features

Note: Attackers use built-in Windows tools and utilities in an attempt to go undetected within the victim environment. This tactic is known as Living Off The Land. Refer to the following resource here to learn more about this.

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

RESULT:

All windows fundamentals were completed successfully.